

ALI YASSER KHUDHAYR

Electrical Engineering, Power & Machines · King Abdulaziz University

Date of Birth: 10 January 2004 **Mobile:** +966 57 033 6515
Nationality: Palestinian **Marital Status:** Single
Residence: Jeddah, Saudi Arabia
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Personal Profile, Career Goal & Vision

Power & Machines engineering candidate with a **4.68 / 5.00 GPA**, skilled in power-system analysis, electrical machines and embedded systems, with real experience in supervision and teaching.

Career goal: a licensed power systems engineer at a leading Saudi energy company.

Vision: a technical leader in Saudi Arabia's renewable-energy sector, serving Vision 2030.

Education

B.Sc. Electrical Engineering — Power & Machines **Jan 2023 – May 2027 (Expected)**
King Abdulaziz University, Jeddah · GPA: 4.68 / 5.00 (Excellent)
Key courses: Power Systems, Protection, High Voltage, Machines, Power Electronics, Control.

Employment History

Research Assistant **8 Oct 2025 – Present**
Center of Excellence in Renewable Energy & Electric Power Systems Research, KAU · Dr. Sultan Alghamdi

- PSO, TDO and hybrid PSO-TDO tuning of a buck-converter PI controller; co-authored manuscript under review.

Field Project Supervisor & Data Reporting (Part-Time) **31 Jul 2026 – 28 Sep 2026**
Coffee and cheese retail projects, Jeddah

- Supervised coffee and cheese projects over two periods; 12 invoice reports and an Excel log for 9 staff at 10 sites.

Teaching Assistant **Aug 2025 – Dec 2025**
Department of Electrical and Computer Engineering, KAU

Organizers Supervisor **Mar 2024 – Feb 2025**
Fitness Club, Deanship of Student Affairs, KAU · supervised 50+ organizers

Industrial Field Visits **Aug 2025 – May 2026**
Al-Taqa Energy · Hala Company · Al-Rowad Factory · Savola Group (Afia)

Projects Involved In

Smart Home Energy Management System (EE499) **Aug 2026 – May 2027**
Senior design: priority-based load shedding and budget tracking; ESP32 baseline at about 910 SAR of a 1,000 SAR budget.

Single-Phase SPWM Inverter Prototype (EE442) **May 2026 – Jun 2026**
12 V DC to 220 V, 60 Hz with Arduino; frequency error below 0.34%.

Quanser QCar 2 Speed & Steering Calibration (EE490) **Aug 2026 – Dec 2026**
Calibration lead: 90-trial Python experiment to cut route error by 30%.

IoT Parking-Slot Detection System (EE366) **May 2026 – Jun 2026**
PIC18 and ESP32 sending occupancy events to a web server.

Mizan: Bookkeeping & Cash-Audit Web App (Freelance, Retail Shop) **Oct 2026**
Offline app replacing a paper ledger, with bank reconciliation and a SHA-256 audit log.

Voluntary Work

Vice President & Treasurer, Hospital Management Club, KAU — **Oct 2026 – Present**
Event Organizer, ECE Club, Sports Hall & KAU-ROBOCON — **May 2023 – May 2025**
200+ Volunteer Hours in education, events and charity — **2019 – Present**

Achievements

- Certificate of Academic Excellence, Deanship of Student Affairs, KAU, 2024–2025.
- Ideathon, Ajyal Care Association: 3rd place (3–4 May 2019); 2nd place (2nd edition, 2022).
- Co-author, "A Comparative Study of PSO, TDO, and Hybrid PSO-TDO Algorithms for Optimal PI Controller Tuning of a DC–DC Buck Converter," submitted to *JKAU: Engineering Sciences* (under review).
- Completed 210+ training hours and earned 50+ certificates.

Charity Work

Volunteer Qur'an Teacher & Supervisor, Umm Al-Qura Mosque — **Jun 2024 – Present**
Ramadan Food Basket Distribution, Ith'am Food Bank — **May 2019 – Jun 2019**

Certificates & Training

IELTS Preparation Program (20 h), IDP IELTS Saudi Arabia — **24 Oct 2026 – 28 Oct 2026**
Introduction to Agentic Programming, GDG, KAU — **5 Aug 2026**
LaTeX Using Overleaf, Writing Center, KAU — **17 Feb 2026**
Qader: Program Design (12 h), Ajyal Association — **26 Jun 2025 – 28 Jun 2025**
Educator Competencies (15 h), Masarat Association — **2 May 2024 – 6 Jun 2024**
Advanced Self-Development (60 h), TVTC — **10 Aug 2022 – 19 Aug 2022**
Microsoft Office 365 (25 h), KAU — **25 Jul 2022 – 11 Aug 2022**

Professional Skills

- **Power:** load flow, fault analysis, protection, machines, power electronics.
- **Software:** MATLAB, Simulink, PSCAD, PowerWorld, LTspice, Proteus.
- **Programming:** Python, C/C++, JavaScript, Verilog; Arduino, PIC18, ESP32.
- **Office:** advanced Excel, Word, PowerPoint, LaTeX.

Soft Skills

- Team leadership
- Teaching and public speaking
- Technical writing
- Planning and goal setting
- Time management

Hobbies

- Designing educational apps
- Motion-graphics videos
- Car mechanics
- Reading and fitness

Languages

Arabic: Native
English: Very good

Other Activities

- 91+ power-engineering explainer videos
- AI-assisted software projects
- Khitatak: offline weekly plan app for Qur'an circles

References

Dr. Fahd Hariri
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Dr. Sultan Alghamdi
EE451 Instructor
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Dr. Omar Alnuqaity
IE201 Instructor
Industrial Eng., KAU

Website *Certificates*